



ABINAYA K

9944818490 | abinayak829@gmail.com | LinkedIn | Github

ABOUT ME

An enthusiastic undergraduate in Artificial Intelligence and Data Science with a strong passion for innovation, problem solving, and the development of intelligent solutions. Eager to contribute technical and analytical skills to impactful projects and grow through meaningful collaborations.

EDUCATION

B.Tech Artificial Intelligence and Data Science

Madras Institute of Technology, Chennai

CGPA 8.74

Aug. 2023 – Present

PROJECTS

Digital Sanitation tracker for organizations-React,Node.js,postgreSQL

- Developed a Digital Sanitation Tracker web application enabling organizations to log sanitation issues, which are routed to workers and monitored by admins in real time. Integrated Socket.IO for real-time updates and built REST APIs to ensure seamless communication across all user roles.

Alumni Database Management System-Flask,Oracle Database

- Developed a web-based Alumni Management System to streamline alumni information management and enhance institutional engagement. Designed the front-end using Flask (Python) for a responsive and intuitive user interface. Implemented a secure and efficient back-end with Oracle Database to manage alumni records, including personal details, career progress, and event participation.

Diet Monitoring System-Flask(python),RESNET18,SVM

- Diet Monitoring System Built a hybrid web application combining CNNs and ML algorithms to detect food items, estimate portions, and calculate calorie intake. Used image preprocessing for segmentation and accuracy improvement. Trained on nutritional datasets to map visual features to real dietary values.

Quantum enhanced Autonomous vehicle system using hybridML and RL-QLearning,Simulation(Ongoing research project)

- Developing a next-generation autonomous driving system that integrates classical reinforcement learning (RL) with quantum-enhanced learning strategies. The project explores how quantum-inspired models can improve decision-making efficiency and adaptability in dynamic driving environments.

SKILLS

- Programming Languages:** Python, C, C++
- Technical Skills:** Machine Learning, OpenCV, MySQL, OracleDB, QISKIT.
- Soft Skills:** Critical Thinking, Problem Solving, Collaboration, Active Listening, Creativity, Leadership.
- Relevant Courses:** Design Thinking – A Primer (NPTEL), Artificial Intelligence Fundamentals(IBM), Data Structures and Algorithms (AU-MIT), Database Management Systems (AU-MIT), Operating Systems (AU-MIT).

ACTIVITIES

- NSS Volunteer:** Contributed to social service projects through the National Service Scheme.
- QUANTuMIT Club Member(Vice President of Technical Activities):** Explored quantum mechanics and computing through interactive sessions and workshops.
- Member of Tamil Mandram (Student club of MIT)**

INTERESTS

- Machine Learning, Deep Learning, Full Stack.
- Problem Solving and Thoughtful Evaluation
- Designing and verbal communication participation

HACKATHONS

Fish Recognition with Chatbot Integration (Agentic Hackathon Finalist)

- Reached the finals of the Agentic Hackathon, showcasing the project's innovation in merging AI-driven vision and conversational AI.